Healthcare Product Management

Module 6: Validating Clinical Fit Before You Build

In healthcare, validation isn't just about user interviews; it's about assessing if a product should exist within clinical and operational reality.

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Course Outline

By the end of Module 6, the you can:

Defend a go / no-go decision for a healthcare product concept using clinical, workflow, data, and safety reasoning.

“Up to now, you’ve designed a product concept.
In this module, you decide whether it deserves to exist.”

The Workflow Compatibility Test

A product that breaks a clinician's workflow is a product that will never be used.

- **"Shadow Workflows":** Understanding the unofficial ways clinicians actually get work done.
- The **"Friction Audit":** Does the tool add steps, clicks, or cognitive load to a busy day?
- **Physical Reality**: Does the tool work in the actual environment (e.g., sterile fields, noisy ERs)?

Artifact Prep: Describe how your product fits into the existing workflow without causing "rejection."

Safety and Liability Exposure

You must validate that your product's risk profile is acceptable to the institution and the law.

- The "Liability Transfer": Does the tool take on clinical risk, or does it leave it all on the clinician?
- Regulatory Validation: Confirming the FDA/CE classification before committing to a roadmap.
- Data Integrity: Validating that the data required for the tool is accurate, timely, and available.

Artifact Prep: Identify the primary liability risk and how the institution will perceive it.

Data Availability Assumptions

Many healthcare products fail because they assume data exists in a format or frequency that it doesn't.

The "Data Gap"

Distinguishing between "Data in the EHR" and "Data accessible via API." Just because it's on a screen doesn't mean it's available for your tool.

Latency Reality

Can your tool get the data fast enough to change the decision in real-time? Batch processing vs. real-time HL7/FHIR feeds.

Quality and Cleaning

Who is responsible for ensuring the input data is "clean" enough? Validating data integrity is a core PM responsibility.

Judgment Call: Technical Reality

Is your product's value dependent on data that is currently "locked" or "dirty"? Remind yourself that "The data is in the EHR" is a dangerous assumption.

Adoption Plausibility: The "Why Now?"

Even a perfect tool will fail if the timing, incentives, or culture aren't ready for it.

Incentive Check

Does the person who has to use the tool benefit from its success? Misaligned incentives are the #1 killer of healthcare tech.

Cultural Readiness

Is the clinical team ready to trust an algorithm or a new digital process? Culture eats strategy for breakfast in the hospital.

Urgency Test

Is the problem painful enough that users will endure the "cost of change"? Status quo is your biggest competitor.

Artifact Prep: Plausibility Argument

List two reasons why adoption might be plausible today versus five years ago. What has changed in the environment to make your product viable now?

Invalidation Criteria: When to Say "No"

A great PM is defined by the products they choose *not* to build.

- Defining "Kill Signals": What specific evidence would prove this concept is non-viable?
- The "No-Go" Logic: If the workflow friction is too high or the data is unavailable, stop.
- Sunk Cost Fallacy: The courage to pivot or kill a project before it consumes more resources.

Artifact Prep: Define three "Invalidation Criteria" that would lead you to stop this project.

Defending Product Viability

To get internal buy-in, you must defend your concept against clinical, technical, and financial skepticism.

- The "Clinical Defense": Proving the tool respects expertise and improves safety.
- The "Technical Defense": Proving the data and integration assumptions are grounded in reality.
- The "Financial Defense": Proving the "Win-Win-Win" for the buyer and the patient.

Judgment Call: Can you look a Chief Medical Officer in the eye and defend this product?

Go / No-Go Decision Checklist

Have you validated that the product is clinically plausible and safe enough to move forward?

Fit to Workflow

- Does this product fit into existing clinical workflows?
- Will it add minimal new burden?

Safety & Liability

- Have specific safety risks been identified?
- Is liability and accountability clear?

Realistic Data

- Is the needed data reliable and accessible?
- Are existing systems compatible with this?

Adoption Plausibility

- Is there a credible champion for this product?
- Can you realistically achieve adoption?

Go or No-Go:

- Doise in all ends-in if..
- Does this product fit into existing clinical workflows?
- Will it add minimal new burden?

No-Go → **Go**

Safety & Liability

- Have specific safety risks been identified?
- Is liability and accountability clear?

Realistic Data

- Is the needed data reliable and accessible?
- Are existing systems compatible with this?

Adoption Plausibility

- Is there a credible champion for this product?
- Can you realistically achieve adoption?

Clear Non-Goals

- Are clear boundaries defined for this product?
- Will users understand what it does NOT do?

Artifact Checkpoint: What You Should Have So Far

Your artifact should now include a rigorous "Clinical Fit & Validation Assessment."

- **Workflow Compatibility**
  Evidence that the tool fits the clinical reality and respects shadow workflows.
- **Safety & Liability**
  A clear-eyed view of the risks, exposures, and regulatory classification.
- **Data Assumptions**
  Validated technical requirements for data access, quality, and latency.
- **Adoption Plausibility**
  A rational argument for why users will change based on incentives and culture.
- **Invalidation Criteria**
  The "Kill Signals" that would lead you to stop or pivot the project.

Summary: Validation as a Mindset

Validation is a continuous process of high-judgment reality checks.

Clinical Fit

Clinical fit is the ultimate "North Star" for healthcare products. If it doesn't align with medical reality, it won't survive.

Data & Workflow

Data availability and workflow compatibility are the most common points of failure. Validate these before you build.

Invalidation

Invalidation is a success, not a failure, if it happens early. A great PM is defined by the products they choose not to build.

Next : Building the Artifact: Finalizing the Healthcare Product Concept Document